MAIS PAC

IPhO 02.12: Special Relativity · Lecture + Problems

Special Relativity

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. Why relativity is needed
2. Time dilation
3. Length contraction
4. Simultaneity, the twin paradox and the pole in the barn
5. Relativistic momentum and energy
6. Problems: reason, calculate, check
7. Further practice and provenance
8. Where to learn more

1. Why relativity is needed

In 1865 Maxwell's equations gave the speed of light as $c = 1/\sqrt{(\varepsilon_0 \mu_0)}$, without saying what it is measured **relative to**. Physicists invented the ether as the medium. Then the Michelson–Morley experiment of 1887 looked for the Earth's motion through it and found nothing: the speed of light was the same in every direction, at every season, to within a tiny fraction.

Einstein's 1905 answer was to take that result seriously and give up something else instead — absolute time.

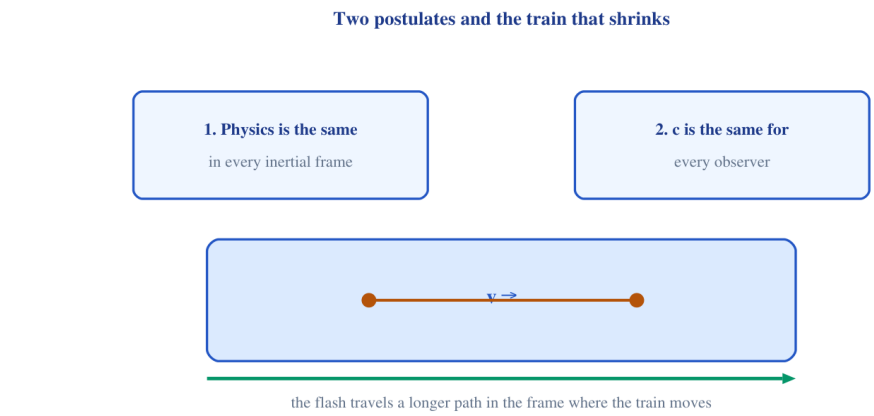


Figure 1. The second postulate is the revolutionary one: the speed of light does not depend on how fast you chase it. Everything else in this topic — time dilation, length contraction, the twin paradox — follows from those two sentences.

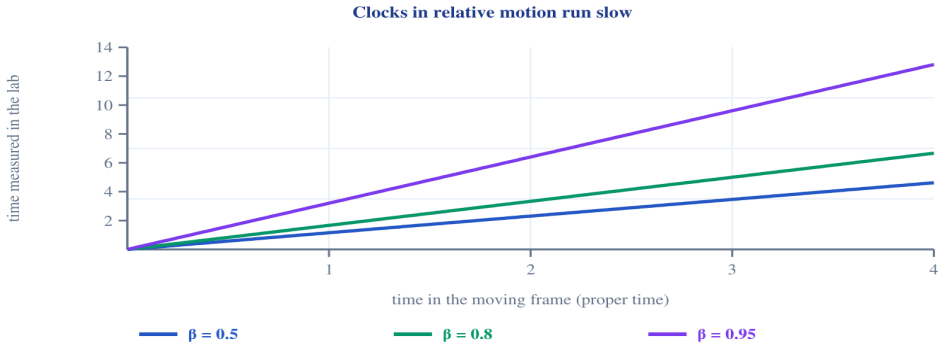
The two postulates. (1) The laws of physics are identical in all inertial frames; no experiment can tell you that you are "really moving". (2) The speed of light in vacuum is the same for all inertial observers, independent of the motion of the source or of the observer.

The second postulate contradicts everyday intuition. If you run towards a light beam at 0.5c, Galileo says the light should approach you at 1.5c; relativity says it still approaches at c. Nothing in the first postulate is surprising; **everything** strange in this topic follows from the second.

2. Time dilation

Consider a light clock: a flash bouncing between two mirrors a distance L apart. In the clock's own frame it ticks every $\Delta t_0 = 2L/c$. In a frame where the clock moves at v, the flash must travel a longer diagonal path, so the tick takes longer:

$$c^2((\Delta t)/(2))^2 = L^2 + ((v\Delta t)/(2))^2 \; \Delta t = (\Delta t_0)/(\sqrt{(1-v^2/c^2)}) = \gamma \Delta t_0. \quad (1)$$



A clock moving at 0.8c ticks once while an identical clock at rest ticks 1.67 times.

Figure. A clock moving at 0.8c ticks once while an identical clock at rest ticks 1.67 times.

The **Lorentz factor** is

$$\gamma = (1)/(\sqrt{(1-\beta^2)}), \; \beta = (v)/(c). \quad (2)$$

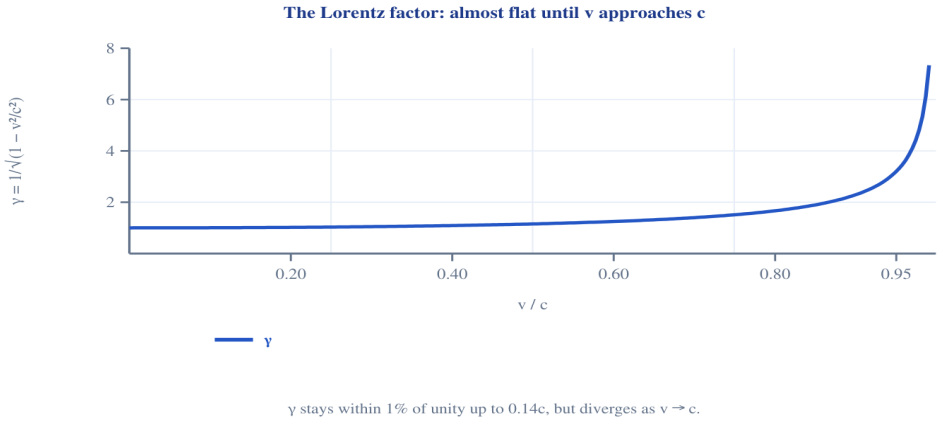


Figure. γ stays within 1% of unity up to 0.14c, but diverges as v → c.

γ = 1 at rest, 1.005 at 0.1c, 1.15 at 0.5c, 2 at 0.866c, 7.1 at 0.99c, and it diverges as v→ c. Below about 0.1c the effect is under 1% — which is why Newtonian mechanics works so well.

The time Δ t_0 measured by a clock present at both events (the clock's own frame) is called the **proper time**, and it is always the shortest time any observer measures between those two events. Muons created in the upper atmosphere live for 2.2 μ{s}, enough to travel only about 650 {m}, yet they reach the ground from 10 {km} up because at 0.998c their lifetime is stretched by γ≈16.

3. Length contraction

The same argument, viewed from the moving frame, gives

$$L = (L_0)/(\gamma), \tag{3}$$

where L_0 is the **proper length**, measured in the object's rest frame. Only lengths along the direction of motion contract; transverse dimensions are unaffected.

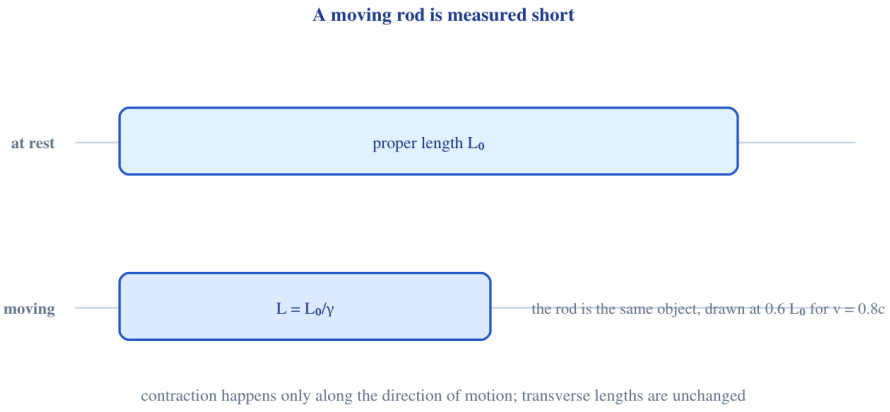


Figure 3. The rod does not squash: the two ends are not measured at the same time in the rod frame, and simultaneity is relative. Length contraction is time dilation seen from the other side.

Time dilation and length contraction are two descriptions of one phenomenon, and neither is "the real one". The muon's journey is 10 {km} long to us and 0.6 {km} long to the muon — both descriptions agree that it reaches the ground.

4. Simultaneity, the twin paradox and the pole in the barn

Two events that are simultaneous in one frame are generally **not** simultaneous in another. In the train thought-experiment, the light from a flash in the middle reaches the front and back of the carriage at different times for a ground observer while it arrives at both ends together for a passenger. That is not a trick of light travel time: it is what "simultaneous" means once c is absolute.

This resolves the classic paradoxes:

- **Twin paradox.** The travelling twin turns round, so she is not in a single inertial frame; the stay-at-home twin is. The asymmetry is physical, and the travelling twin really does come back younger. The effect is measurable with atomic clocks on aircraft and satellites (GPS would drift by kilometres a day without correction).
- **Pole in the barn.** A long pole moving fast fits inside a short barn as seen from the barn frame, but in the pole frame the barn is contracted and the pole does not fit. The resolution is that the two doors are never closed simultaneously in the pole frame.

5. Relativistic momentum and energy

To keep momentum conserved in all frames, momentum must be redefined as

$$p = \gamma\, m\, v, \tag{4}$$

and Newton's second law becomes F = dp/dt. The work done in accelerating a mass therefore gives a kinetic energy

$$K = (\gamma - 1)mc^2, \tag{5}$$

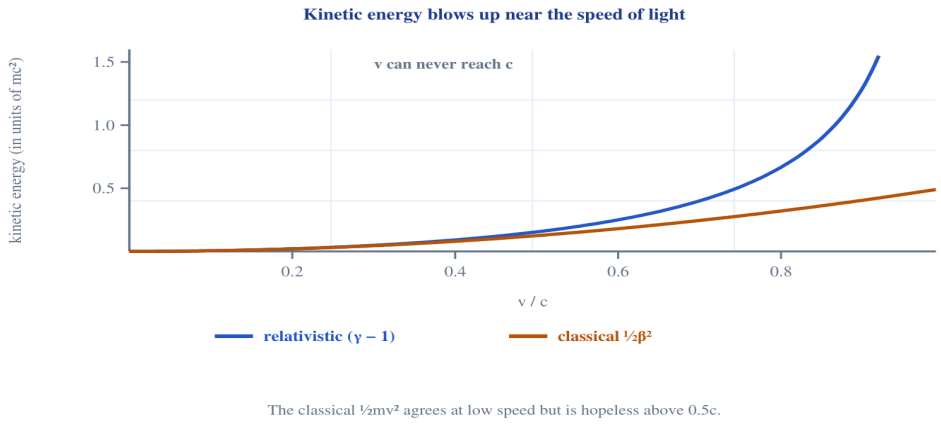


Figure. The classical ½mv² agrees at low speed but is hopeless above 0.5c.

which as v≪ c reduces to (1)/(2)mv². The total energy is

$$E = \gamma\, mc^2,\ E^2 = (pc)^2 + (mc^2)^2. \tag{6}$$

The second form is worth memorising: it holds for massless particles too. For a photon, m=0 gives E = pc, which is why light carries momentum despite having no mass — the basis of solar sails and of Compton scattering.

Rest energy. Setting v=0 gives E_0 = mc²: mass **is** energy. One gram is 9×10^(13) {J}, about the yield of a small nuclear weapon, and the Sun converts 4×10^9 {kg} of mass per second into light. In nuclear reactions the mass difference between reactants and products appears as kinetic energy, and the fractional mass change is tiny but measurable: for the deuterium–tritium fusion, about 0.4%.

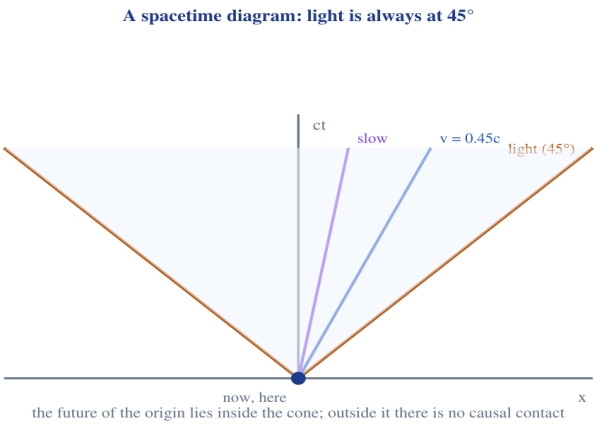


Figure 5. Because every inertial observer agrees on c, the light rays are the same 45° lines in every frame; that is what forces the Lorentz transformation, and with it the loss of absolute simultaneity.

Interactive experiment: Relativistic travel — open the online handout to use it.

The relativity checklist. Which frame is the proper frame? (the one in which the two events happen at the same place for time, or in which the object is at rest for length) → compute γ first, then everything else is multiplication or division → proper time is the shortest, proper length the longest → never add velocities naively (the relativistic addition is u' = (u-v)/(1-uv/c²)) → for energy problems, use E² = (pc)²+(mc²)² and remember K = (γ-1)mc² → if the answer involves v>c, something has gone wrong in the algebra, not merely in the physics.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — The Lorentz factor. A spacecraft travels at 0.60c relative to Earth. (a) Find γ. (b) Find the factor by which its clocks run slow and its length contracts. (c) Find the speed at which γ = 2.

Worked solution →

[1] Problem 2 — Muons from the upper atmosphere. Muons are created 10 {km} above the ground with a speed of 0.998c and a proper lifetime of 2.2 μ{s}. (a) Find γ. (b) Find the distance a muon travels in the Earth frame before decaying. (c) Explain the same survival in the muon's own frame, using length contraction.

Worked solution →

[2] Problem 3 — A trip to a nearby star. A star is 4.3 light years away. A spacecraft travels there at 0.80c. (a) Find the time taken in the Earth frame. (b) Find the time elapsed on the spacecraft's clock. (c) Find the distance travelled in the spacecraft's frame, and check that the two descriptions agree.

Worked solution →

[2] Problem 4 — Relativistic kinetic energy. A proton has rest mass 1.67×10^(-27) {kg}. (a) Find its total energy at 0.90c. (b) Find its kinetic energy in MeV and compare with the classical value. (c) Find the momentum at that speed and check the relation $E^2 = (pc)^2+(mc^2)^2$.

Worked solution →

[2] Problem 5 — Mass into energy. A nuclear power station produces 1.0 {GW} of electrical power with an efficiency of 35%. (a) Find the mass converted to energy per second in the reactor. (b) Find the mass lost per year. (c) Explain why the answer is small and why it nevertheless matters.

Worked solution →

[2] Problem 6 — Velocity addition. A spacecraft moves at 0.80c away from Earth and fires a probe forwards at 0.60c relative to the spacecraft. (a) Find the speed of the probe in the Earth frame using the relativistic formula. (b) Compare with the Galilean answer and explain the difference. (c) Show that the relativistic formula never gives a speed greater than c when both speeds are below c.

Worked solution →

[2] Problem 7 — The twin paradox. One twin stays on Earth while the other travels at 0.90c to a star 9.0 light years away and returns at the same speed. (a) Find the total time on Earth. (b) Find the time experienced by the travelling twin. (c) Explain why the situation is not symmetric and why the traveller really is younger.

Worked solution →

[3] Problem 8 — A relativistic particle in a magnetic field. An electron moves at 0.95c in a magnetic field of 1.0 {T} perpendicular to its velocity. (a) Find γ and the relativistic momentum. (b) Find the radius of its path, using r = p/(qB). (c) Compare with the classical radius and explain why accelerators must be designed relativistically.

Worked solution →

[3] Problem 9 — The lifetime of a particle beam. A beam of unstable particles with proper lifetime 1.0×10^(-8) {s} is produced at 0.99c. Half of them decay within how long, and is the laboratory distance travelled equal to γ times their rest-frame distance? (a) Find γ. (b) Find the laboratory lifetime and decay length. (c) Explain the statement "the particle sees the laboratory contracted" and check it gives the same length.

Worked solution →

[3] Problem 10 — Energy needed to reach a speed. A 1000 {kg} spacecraft is to be accelerated from rest to 0.90c. (a) Find the kinetic energy required, in joules and in units of the rest energy. (b) Find the total energy of the spacecraft at that speed. (c) Explain why the energy cost grows without limit as v→ c and what this implies for interstellar travel.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

PHYSNET / modern physics overview

> Short modules on quanta, spectra, relativity and nuclear physics written for students.